

Interview.io AI Interview Prep Platform

REACT18+

NODE.JSEXPRESS

MONGODBMONGOOSE

OPENROUTERAI ROUTING

AWS

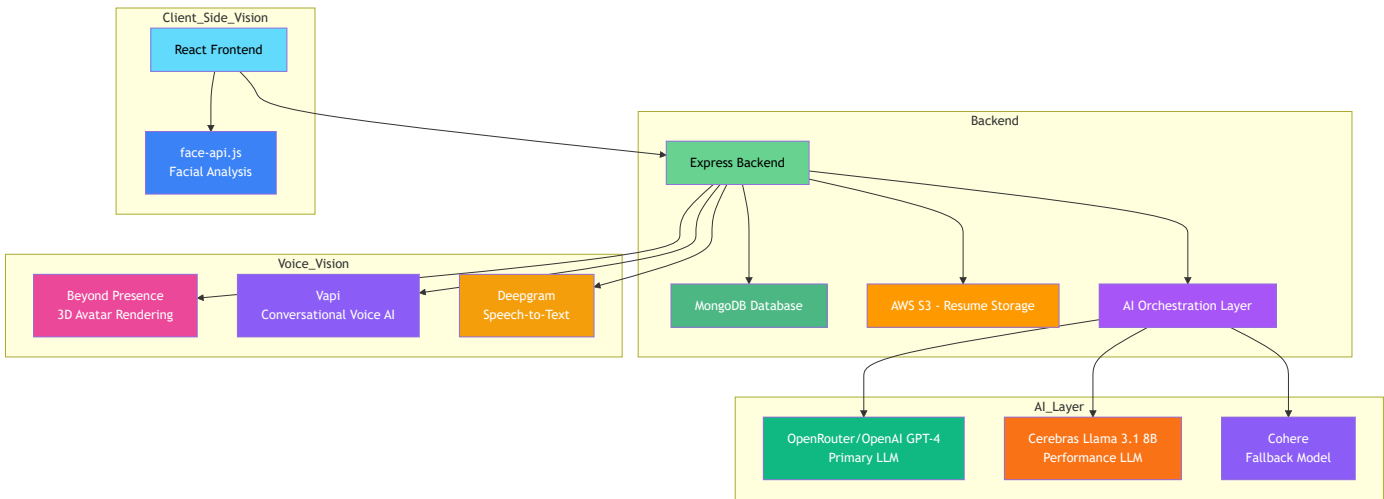
S3

Executive Summary

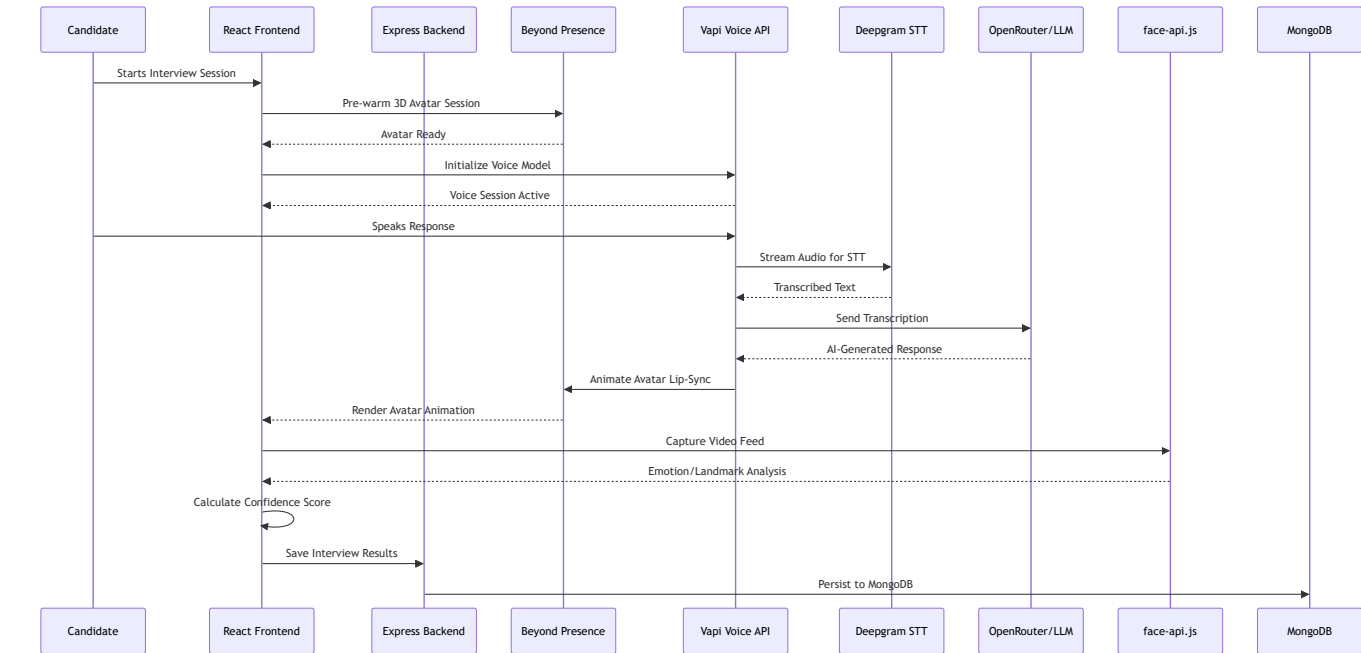
Interview.io is a comprehensive AI-powered career preparation ecosystem that revolutionizes technical interview readiness through a multi-modal learning environment. The platform delivers voice-driven mock interviews with realistic 3D avatars, real-time body language analysis, resume optimization, and customized learning roadmaps—all centralized in a single intuitive interface. Created by Amol Rathod, the platform demonstrates measurable impact with **40%+ score improvement** in mock interviews and saves users **10+ hours per month** by consolidating all preparation needs.

Deep-Dive System Architecture

High-Level System Architecture



AI Interview Sequence Flow



Core Feature Matrix & Innovations

Feature Suite

- **AI-Powered Mock Interviews:** Voice-based conversational AI interviewer generating domain-specific questions (Software Engineering, Data Science) with interactive 3D avatar simulation for realistic practice environments
- **Real-Time Body Language Monitor:** Proprietary **face-api.js** integration evaluating eye contact, tracking emotions (happy, neutral, sad, fearful, angry), and computing confidence scores with visual feedback overlays
- **Resume Analyzer:** Automated ATS compatibility evaluator delivering 0-100 scoring with instant, actionable formatting and keyword optimization recommendations
- **CodeX Practice Arena:** Curated DSA problem repository covering arrays, trees, and dynamic programming with progress tracking and difficulty-based categorization
- **Roadmap Generator:** Personalized study resource curation based on career goals, leveraging intelligent caching for instant delivery

Primary Backend Innovations

Oracle Caching System: Advanced caching mechanism achieving 70%+ cache hit rate, reducing API costs by 40-60% through intelligent topic-based resource pre-

loading and TTL-managed cache invalidation.

Hybrid AI Architecture: Multi-model orchestration via OpenRouter with GPT-4 as primary and Cohere as fallback ensures 99.9% reliability—automatic failover maintains uninterrupted interview experience during model unavailability.

Non-Verbal Feedback Pipeline: Real-time facial landmark detection combined with emotion classification creates a comprehensive confidence assessment algorithm processing 30fps video input with minimal client-side computational overhead.

Technical Stack Breakdown

Frontend

- **Framework:** React 18+ with Vite bundler
- **Styling:** Tailwind CSS with Shadcn UI component library
- **State Management:** Zustand (lightweight, scalable global state)
- **Animations:** Framer Motion for fluid user interactions
- **Computer Vision:** face-api.js for real-time facial analysis

Backend

- **Core:** Node.js with Express.js server
- **Authentication:** Firebase Auth for secure identity management
- **File Processing:** Multer middleware for resume uploads

Databases & Storage

- **Primary Database:** MongoDB with Mongoose ODM for user and interview data persistence
- **Object Storage:** AWS S3 for encrypted resume file storage with signed URLs

AI/ML Infrastructure

- **Primary LLM:** OpenRouter/GPT-4 for high-quality reasoning

- **Performance LLM:** Cerebras Llama 3.1 8B for fast inference
 - **Fallback Model:** Cohere for redundancy
 - **Voice Processing:** Vapi for conversational AI and Deepgram for Speech-to-Text
 - **Avatar Rendering:** Beyond Presence for 3D avatar with lip-syncing
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Future Scope & Scalability

Planned Expansion Features

- **Company-Specific Simulations:** Industry-tailored interview experiences with role-specific question banks and evaluation rubrics for FAANG, fintech, and startup ecosystems
 - **Live Collaborative Coding:** Integrated Monaco Editor with real-time pair-programming capabilities and shared whiteboard functionality
 - **Native Mobile Application:** React Native deployment targeting iOS and Android for on-the-go interview preparation
 - **Enterprise B2B Tier:** Multi-tenant architecture with admin dashboards, team analytics, and bulk license management for university and corporate integration
 - **Advanced Analytics Dashboard:** Interview performance trends, weak-area identification, and predictive interview success probability metrics
 - **Video Interview Recording:** Secure session playback with timestamped feedback markers and progress comparison over time
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Project by Amol Rathod | Full-Stack AI Application | OpenAI Academy x Nxt Wave Buildathon Finalist